Mian (Matt) Si

Software Engineer · Backend & Full Stack · Java, Python, Ruby on Rails · Applied AI

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SUMMARY

- Software engineer with **15+ years** building enterprise, financial and SaaS-style systems for banks, trading desks and logistics operations at Moody's Analytics, Ping An Securities and S.F. Express.
- **Fixes root causes, not symptoms:** replaced brute-force cycle detection with a topological sort and cut a production tool's startup **from hours to seconds**; tuned a Rails query from **358 ms to 0.4 ms** with a composite index.
- **Backend to full stack:** Java and Python services for trading and logistics; recently built and deployed **CaseDesk**, a multi-tenant web app (Rails 8, Hotwire, MySQL, CI, Docker, Kamal).
- **Applied AI:** machine learning specialist on an applied computer vision research project (PyTorch, Detectron2); builds personal projects with **Claude Code**, reviewing every change through pull requests and CI.
- **Mentors engineers** through domain walkthroughs, codebase tours and code review; works well with distributed teams and clients across China, the US and Canada.

TECHNICAL SKILLS

Languages: Java, Python, Ruby, C#, C++, SQL

Web & backend: Java services, Ruby on Rails 8, Hotwire (Turbo, Stimulus), Tailwind CSS, web services (WCF), authentication, batch processing

Databases: MySQL (query plans, composite indexes, benchmarking at 1M rows), SQL Server (high-availability mirroring)

AI / ML: PyTorch, Detectron2, torchvision, object detection, Whisper speech recognition, neural TTS; AI-assisted development with Claude Code

Testing & delivery: Minitest, RuboCop, Brakeman, Selenium, SpecFlow (BDD), GitHub Actions CI, Docker, Kamal, Git and pull requests

Practices: Agile delivery, code review, mentoring, requirements analysis, production support

EXPERIENCE

Machine Learning Specialist — Sheridan College — Oct 2026 – Present

Brampton, ON · Hybrid · Contract

- Apply computer vision and deep learning to automated inspection in aerospace as part of an applied research team.
- Develop and evaluate deep learning models that detect defects in inspection imagery, and turn experiment results into practical inspection workflows with the research team and industry stakeholders.

Research Assistant — Sheridan College (Mitacs Business Strategy Internship) — Dec 2025 – May 2026

Toronto, ON

- Researched computer vision and deep learning methods for an industry-sponsored visual inspection project.
- Ran experiments, analyzed model results and presented findings to researchers and the industry partner.

Senior Software Engineer, Backend & Test Automation — S.F. Express — Apr 2021 – Mar 2023

Shenzhen, China

- Built features and upgrades for **internal enterprise logistics systems** at one of China's largest logistics companies, in **Java and Python**.
- Built and maintained **automated test frameworks** that improved release quality and reduced manual testing.
- Turned complex business requirements into technical tasks and worked with business, development and QA teams to **troubleshoot production issues** and improve reliability.
- **Mentored new engineers:** walked them through the logistics domain and the codebase module by module, and reviewed their code until they delivered independently.

Senior Software Engineer — Ping An Securities — Jun 2017 – Apr 2021

Shenzhen, China

- Developed and maintained **Java fixed-income trading and analytics systems** used by trading and risk teams.
- **Converted Excel-based bond and interest-rate swap valuation and risk models into production services**, covering cash-flow processing, P&L calculation and batch analytics.
- Built **Python reconciliation tools** that checked end-of-day results automatically and reduced manual checks.

- Provided production support and delivered enhancements with trading, risk and business stakeholders.

Senior Software Engineer — Moody's Analytics Jun 2015 – Apr 2017
Shenzhen, China

- Implemented client-specific internal credit rating models for **major banks** in RA Studio and C#, turning approved credit-risk specifications into production software.
- **Performance root-cause fix:** replaced brute-force cycle detection in DAG-based rating models with a **topological sort**, cutting model-viewer startup **from hours to seconds**.
- Helped build a **Selenium + SpecFlow (BDD) automation framework** with QA and business analysts, covering about ten scorecards.
- Onboarded and mentored new team members on the credit-risk business and the rating-model modules, and reviewed their code.

Senior Software Engineer — One Cloud Sep 2013 – Jun 2015
Guangzhou, China

- Developed modules for **private cloud products** and designed and implemented their **authentication flows**.
- Designed a **high-availability solution based on SQL Server mirroring**.
- Built a content and rules publishing module so customers could update content and rules without a release.

Earlier experience 2007 – 2013

- **Software Engineer, Pactera Technology** (client: Expedia), Shenzhen & Seattle, 2012–2013: hotel inventory web services; refactored core services from C++ to WCF (C#); built a web tool for test managers.
- **Software Engineer, Moody's Analytics**, Shenzhen, 2008–2011: credit rating scorecards for banking clients in C#; contributed to the RiskAnalyst platform and RA Studio.
- **Software Developer Intern, Thomson Reuters**, 2007–2008: developed and maintained the indexing module of a **legal search engine**.

SELECTED PROJECTS

Full write-ups, screenshots and live demos at mattsi.github.io

CaseDesk — Ruby on Rails 8, Hotwire, MySQL 8, Tailwind CSS, Kamal · Live demo · GitHub

- Multi-tenant case management app for law firms: a Today page with overdue tasks first, then the next seven days of tasks and events grouped by day.
- Every lookup is scoped to the signed-in user so firms never see each other's data; per-firm time zones decide what counts as "today"; Turbo Streams update the page without a reload.
- **Database optimization:** benchmarked with 1M tasks and 200K events, read the query plans and added a composite index on (assignee_id, status, due_at): Top priority **358 ms → 0.4 ms**, Upcoming **377 ms → 4.1 ms**.
- Minitest, RuboCop and Brakeman on GitHub Actions; deployed with Docker and Kamal to Google Cloud.

Steel Defect Detection — Python, PyTorch, Detectron2 · GitHub

- Two-stage detector (Faster R-CNN finds defects, ResNet-50 names them) compared with a single six-class baseline on NEU-DET: COCO AP **0.401 vs 0.391**, 22 more defects found; all thresholds chosen on validation only.

ListenStudio — Python, faster-whisper, PyAV, neural TTS · GitHub

- Desktop app for English listening and speaking practice, built with **Claude Code** as a pair programmer through pull requests and CI: background transcription with pause and cancel, word-by-word pronunciation checks, Windows and macOS builds in GitHub Actions.

OpenGL Rendering Engine — C++17, OpenGL 3.3, GLSL · GitHub

- Real-time 3D renderer from scratch: component-based scene system loaded from JSON, GPU-side procedural instancing and a chain of ten post-processing passes.

EDUCATION

Graduate Certificate, Game Development – Advanced Programming — Sheridan College 2025 – 2026

Graduate Certificate, Internet of Things & Machine Intelligence — Sheridan College 2024 – 2025

M.Eng., Software Engineering — Peking University 2005 – 2008

B.Eng., Computer Science & Technology — Anhui Polytechnic University 2000 – 2004

LANGUAGES

English (professional working) · Mandarin Chinese (native) · French (limited working)